

Lab 3 — Architectural Pattern Selection & Justification

Blood Bank Inventory & Emergency Donor Matcher

We chose Microservices Architecture for the Blood Bank Inventory & Emergency Donor Matcher System.

Reason 1 — Independent scaling for emergency alert bursts

NFR-002 requires the system to remain responsive under up to 20x normal request volume during mass-casualty events. In a microservices architecture, the Donor Matching Service and Notification Service — the two components that see the biggest spike in load during an emergency — can be scaled out independently, while lower-traffic services like Donor Registry stay at baseline. This targeted scaling avoids over-provisioning the entire system just to handle a spike in one workflow.

Reason 2 — Fault isolation between critical and peripheral functions

SMS gateway integrations (used by the Notification Service) are a common source of third-party outages or slowdowns. Because each service is independently deployable, a failure or slow response in the Notification Service does not block the Emergency Request Service from accepting new requests or the Inventory Service from maintaining accurate stock. This isolation keeps the system's core intake and inventory functions available even if a peripheral integration degrades.

Security Advantage

Each microservice can enforce its own least-privilege access boundary. Only the Inventory Service holds direct write access to the blood-stock ledger, which lets it enforce the strict transactional consistency required by NFR-001 (preventing double-allocation of a bag). A bug or compromise in the Donor Matching or Notification services cannot directly manipulate stock counts, since all cross-service communication happens through defined, independently authenticated and rate-limited APIs rather than shared database access.

Performance Benefit

The Donor Matching Service is the most compute-intensive path in the system, since it cross-references blood-group compatibility, geolocation radius, and donor eligibility in real time. Because it is a separate, independently scalable service, additional instances can be spun up specifically for this workload during high-demand periods without needing to scale the rest of the system. This targeted scaling is what keeps the 30-second donor-notification SLA in FR-001 achievable even under peak load.